

LIGHT-WEIGHT POWER SUPPLIES Are Not Just Benchtop Anymore



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In case of power supplies, the move from a wall outlet to a benchtop might not be a significant development when we have scientists working on wireless transmission of electricity. However, the move to portable power supplies has seen some interesting trends of its own. A lot of focus is being laid on the supplies to be operable in constant current, constant voltage, constant resistance, constant power and constant impedance modes.

Advances have also been in the fields of current measurement with nano-amp resolution, variable rise and fall times, digital input-output, multiple PC interfaces, battery-simulating capabilities and kilovolt outputs.

Raghunath Rao K., manager application team, Tektronix India Pvt Ltd, says his company is focusing on minimising power consumption to increase battery life for battery simulators. He explains, "You need a source measure unit with low current-measurement sensitivity to evaluate and qualify these devices for your design." The source measurement unit can test a device over the voltage range of the battery.

Sumit Sharma, marketing manager, Good Will Instrument Co. India, says, "One particular requirement of power consumption for portable wireless communications devices is pulse current." Pulse current measurement indicates transient current consumption, which helps in al-

location of the power supply system for a product's preliminary design, that is, power supply circuits and battery selection for product's product analysis.

Today, the power supply industry is at the beginning of a major focus shift that puts efficiency improvements across the entire load range at the forefront of customers' performance requirements. "Newer supplies offer resolutions as low as 1mA and 1mV," adds Praveen Kumar Gupta, director, Avante Global Services Pvt Ltd. "Power supplies with the ability to measure very low currents," says Rao, when asked about development areas.

PPH-1503 from GW Instek provides 0.1mA (equivalent to 4% digits) current resolution under 5A setting. "It is capable of rise-time of 0.15 minutes and fall-time of 0.65 minutes," says Sharma.

Multiple channel output. A common trend has been to increase the number of output channels. Though the most common type of supply remains the two-channel, we also have multiple channels coming in. MDL series from B&K Precision is one such example. The series offers support for up to four modules or eight channels, with the number increasing to up to eight modules or 16 channels with mainframe extension.

Similar trends are seen in other cases as well. The 382270 quad-output DC power supply from Etech has two 0 - 30V at 0 - 5A outputs plus two fixed auxiliary outputs of 3V - 6.5V at 3A and 15V at 1A, taking the technical output count to six channels.

Global Specialties has launched the 1415 dual-output DC power supply. It features up to 30V at 3A with dual independent outputs, wherein one fixed output produces 5V at 3A. Load condition changes with automatic crossover between constant current and constant

Fig. 1: A precision measurement DC supply

